

the lateral reticulospinal tract. Significance: It has no intrinsic activity.

5. Stretch reflex (II)

Clinical Applications of the Stretch Reflex

(1) The dynamic response:

A- The tendon jerk (figure 22, 23):

- Definition: Sudden stretch of the muscle by tapping on its tendon → reflex rapid contraction of the muscle followed by rapid relaxation.
- Mechanism: Sudden stretch of the muscle → stimulation of primary endings of the muscle spindle → ↑ afferent discharge from the spindle → enter the spinal cord along the dorsal roots and end directly on the αAHCs → ↑ efferent discharge to extrafusal muscle fibers → reflex contraction and shortening → ↓ or stop afferent discharge from the spindle → muscle relaxation.

Examples (table 9):

Reflex	Description	Center
The biceps reflex	Tapping on the biceps tendon → reflex contraction of the biceps and flexion of the elbow.	C4- C5
The triceps reflex	Tapping on the triceps tendon → reflex contraction of the triceps the and extension of elbow	C5 - C6
The knee jerk	Tapping on the patellar tendon → contraction of the quadriceps and extension of the knee	L3 - L4
The ankle jerk	Tapping on the Tendon Achillis → reflex contraction of the muscle gastrocnemius and plantar flexion of the ankle	S1- S2

Neuro 25-26

Significance (table 10:

Lost reflex (areflexia)	Weak reflex (hyporeflexia)	Exaggerated reflex (hyper- reflexia)
Occurs when the reflex arc is injured at any facilitatory point.	Occurs when supraspinal facilitatory centers or their descending tracts are affected → dominant inhibition.	Occurs when supraspinal inhibitory centers or their descending tracts are affected → dominant facilitation.
Associated with lost muscle tone (aotonia)	Associated with weak muscle tone (hypotonia)	Associated with exaggerated muscle tone (hypertonia)

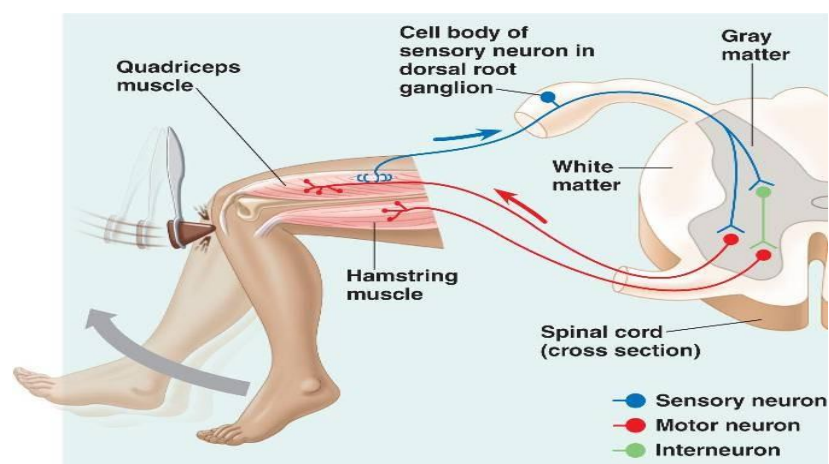


Figure 22: the knee jerk



Biceps Jerk



Triceps Jerk



Knee Jerk



Ankle Jerk



Figure 23 tendon jerks



Neuro 25-26

B- Clonus:

Definition: It is oscillation of the tendon jerk when the muscle tendon is:

* stretched and * the stretch is maintained on the condition that the stretch reflex is

- facilitated (hyper-reflexia) and there is
- hypertonia.

facilitated and not under normal condition.

Mechanism:

- Sudden stretch of the muscle → stimulation of primary endings of muscle spindle → ↑ afferent discharge from the spindle → enter the spinal cord along the dorsal roots and end directly on the on αAHCs → ↑ efferent motor discharge to extrafusal muscle fibers → reflex contraction and shortening → ↓ or stop afferent discharge from the spindle → muscle relaxation, but since the stretch is maintained → reflex contraction occurs once more and the cycle repeats itself.

Examples (table 11):

Reflex	Description	Center
The knee clonus	Sudden maintained stretch of the patellar ligament → reflex oscillations of contractions and relaxations of the quadriceps muscles.	L3- L4
The ankle clonus	Sudden maintained stretch of the Tendon Achillis by ankle dorsiflexion → reflex oscillations of contractions and relaxations of the gastrocnemius muscle.	S1 -S2

Neuro 25-26

(2) The static response:

- The static component of the stretch reflex is the muscle tone.
- The mechanism and functions of muscle tone were previously discussed.
- Variations in muscle tone with different diseases was mentioned before with the tendon jerk.

Properties of the stretch reflex

1. Localization: highly localized, the only stretched muscle contracts.
2. Irradiation: increased stretch $\uparrow$ degree of muscle contraction.
3. Central delay : very short as it is monosynaptic.
4. Fatigue: occurs with repeated stimulation but of slow onset because:
 - * The afferent impulse frequency is low $\rightarrow$ incomplete tetanus $\rightarrow$ allows for partial relaxation.
 - * The motor units alternate in function; some are contracting while the others are relaxing $\rightarrow$ maintained tension in the muscle.
 - * Partial relaxation allows for better oxygenation.
1. Reciprocal inhibition: Reflex contraction of a muscle by stretch is associated with relaxation of the antagonist.

The Inverse stretch reflex (lengthening reaction) (figure 24)

- Definition : Severe increased muscle tension as in :
 1. forcible contraction against resistance, or
 2. severe passive stretch $\rightarrow$ reflex feedback inhibition (relaxation) of the muscle $\rightarrow \downarrow$ muscle tension.

Neuro 25-26

- Pathway:
 1. Stimulus: $\uparrow$ muscle tension.
 2. Receptor: Golgi tendon organ in the muscle tendon. Afferents: rapidly conducting thick myelinated A_B fibers. Center: polysynaptic; the afferents end on inhibitory interneurone secreting glycine that inhibit the α AHCs of the severely stretched muscle.
 3. Response: muscle relaxation (inhibition).
 4. Significance: Prevents * rupture of the muscle, or * avulsion of the tendon.

N.B.

- The stretch reflex $\rightarrow$ feedback reflex maintaining the muscle length.
- The inverse stretch reflex $\rightarrow$ feedback reflex maintaining the muscle tension.

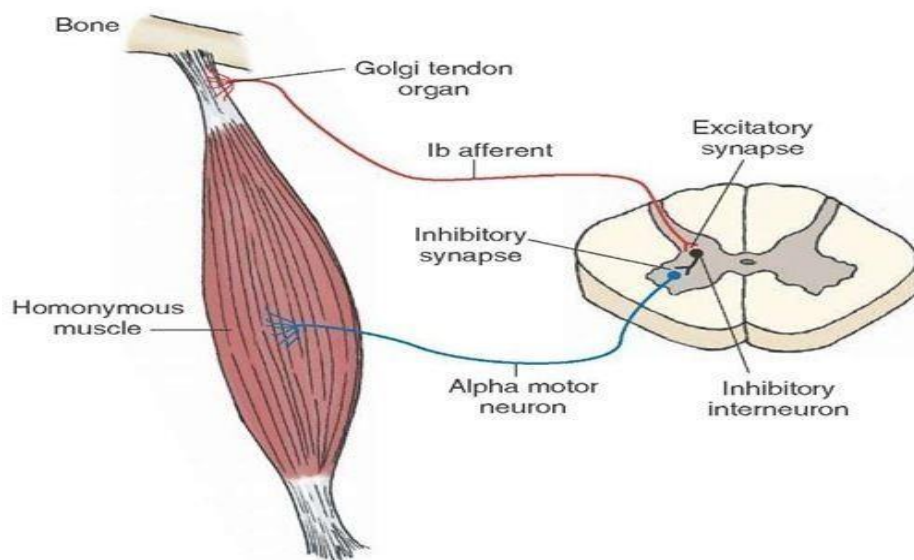


Figure 24 : Inverse SR

Polysynaptic spinal cord reflexes:

(A) Somatic reflexes

(1) Flexion withdrawal reflex (figure 25):

- Definition: Application of an injurious stimulus to a limb → - reflex contraction of the flexors to get the limb away from the injurious stimulus.
- Significance: protection.
- Properties: The reflex shows:
 1. Localization
 2. Irradiation.
 3. Central delay. > 2 ms.
 4. Reciprocal inhibition.
 5. After discharge: The flexors remain contracted for sometimes after getting away from the injurious stimulus.
 6. Fatigue, block and resistance.
 7. Prepotent: inhibits all other reflexes occurring at the same time e.g. scratch reflex.

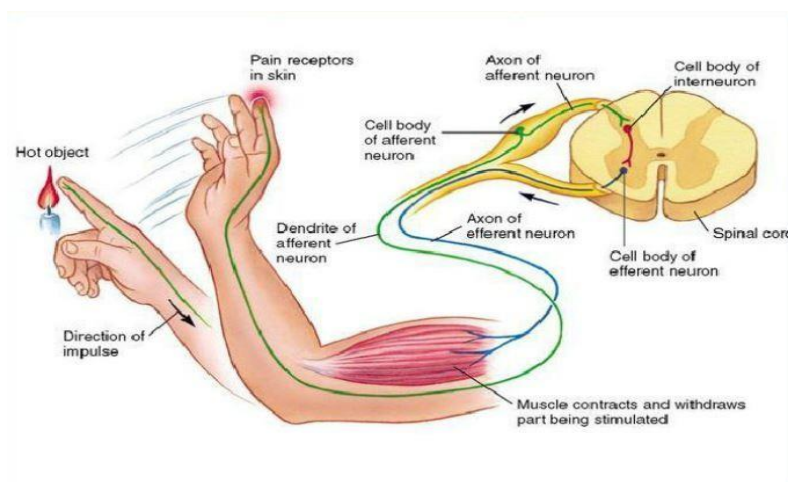


Figure 25: Flexion withdrawal reflex

(2) Postural spinal cord reflexes

Neuro 25-26

A) The crossed extensor reflex (figure 26):

- Definition: Application of an injurious stimulus to a limb → Flexion withdrawal of the stimulated (ipsilateral) limb, while the extensors of the contralateral (opposite) limb contract to support the body weight (crossed extensor reflex).
- Properties: The reflex shows: 1- Localization, 2- Irradiation, 3- Central delay (long) → polysynaptic 4- Reciprocal inhibition, 5- After discharge: The extensors remain contracted after getting away from the injurious stimulus.

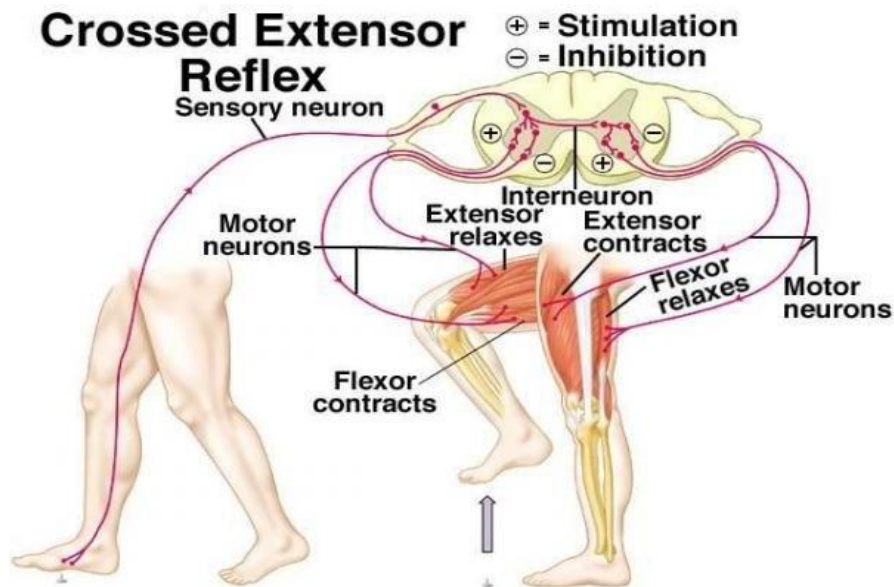


Figure 26: The crossed extensor reflex

B) Positive supporting reflex:

- Definition: Application of deep pressure to the sole of foot → reflex contraction of both flexors and extensors of the lower limb → the lower limb becomes a solid pillar to support the body weight during standing.
- Properties: The reflex shows all reflex properties except reciprocal inhibition.

Neuro 25-26

C) Stepping reflex of a single limb:

Definition: Forward flexion of a single limb is followed by backward extension and the cycle repeats itself during walking. It is due to reciprocal inhibition and rebound.

D) Reciprocal Stepping reflex of the opposite limb:

- Definition: During walking, Forward flexion of one limb is associated with backward extension of the contralateral limb.
- Significance: The extended limb supports the body weight while the flexed limb is taking off the ground.

E) Diagonal stepping of all four limbs:

Definition: During walking, Forward flexion of the right forelimb and left hindlimb occurs while the left forelimb and right hindlimb are in backward extension and the cycle alternates.

(3) Superficial skin reflexes:

1. Upper and lower abdominal reflexes:

Definition: Stroking the skin of the abdomen → reflex contraction of the abdominal muscles and shifting of the umbilicus towards the site of the stimulated skin (table 12).

Table 12: abdominal Reflexes

Site of stimulated skin	Center
Above the umbilicus	$T_7 - T_8 - T_9$
At the level of umbilicus	T_{10}

2- The cremasteric reflex:

- Definition: Stroking the skin of the inner aspect of the thigh → reflex contraction of the cremasteric muscle and elevation of the testicles.

Neuro 25-26

- The center is S₁– S₂.

2- The planter reflex (Figure 27):

- Definition: stroking the skin of the sole of the foot → reflex planter flexion of the toes.
- The center is S₁-S₂.



Figure 27: The planter reflex

(B) Autonomic reflexes

- 1) The micturition reflexes.
- 2) The defecation reflex.
- 3) The vascular reflexes in response to skin temperature changes:
 - Application of hot object to skin → reflex vasodilatation. *
 - Application of cold object to skin → reflex vasoconstriction. *
 - Application of hot object to skin → reflex sweating.
- 4) Irritation of the peritoneum → reflex inhibition of gastrointestinal motility.